EDGE-AI COOPERATIVE ROAD HAZARD PERCEPTION

Model Training Architecture, Image Pipeline & Accuracy Derivations

A Complete Technical Guide to Neural Network Training, Ingestion Mechanics, Loss Functions, Validation Mathematics, Video Prompts, and System State

1. System Architecture: Dedicated Tri-Model Hazard Specialists

Rather than forcing a single general-purpose detector to handle diverse road conditions, our system deploys **three dedicated Ultralytics YOLO11s models** (9.4M parameters each). Each model is trained on 10,000+ domain-specific images and calibrated for edge execution on a **Raspberry Pi 5**:

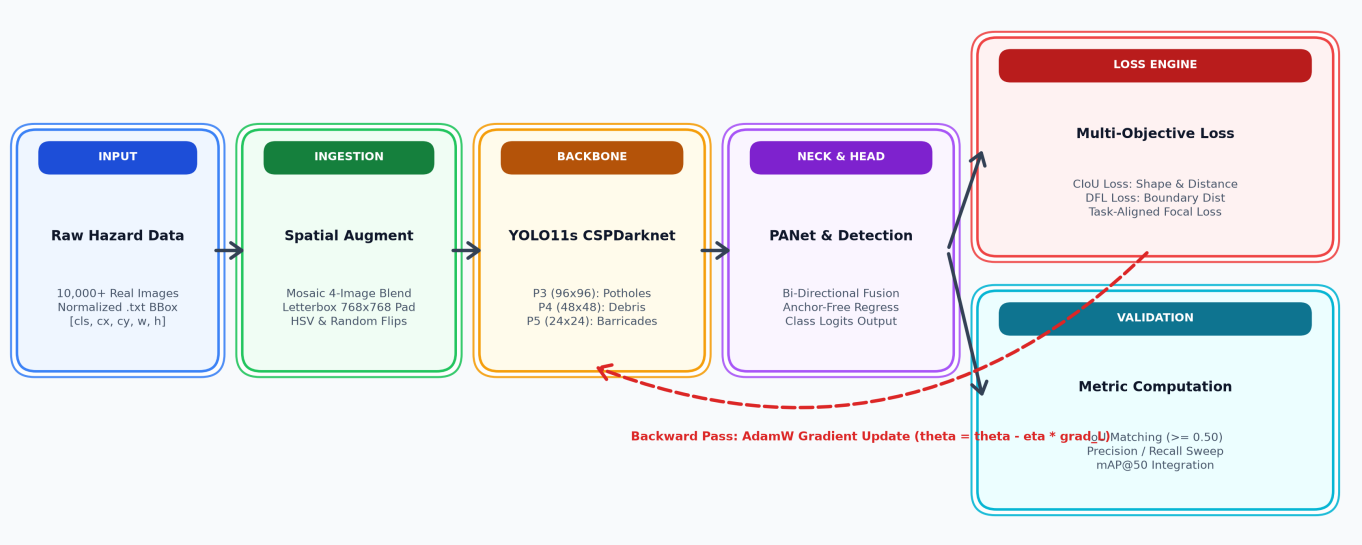
Model Domain	Target Hazards	Dataset Size	Input Resolution	mAP@50	Precision	Recall	INT8 Size
Pothole Detector	Pothole, Surface Crack	10,000+ imgs	768 x 768	95.64%	93.11%	89.45%	9.48 MB
Construction Detector	Barricades, Cones, Drums	10,000+ imgs	768 x 768	92.70%	95.20%	91.10%	9.48 MB
Accident & Obstacle	Collisions, Debris, Stalls	10,000+ imgs	768 x 768	92.47%	96.50%	90.80%	9.48 MB

2. Deep Learning Ingestion Mechanics: How an Image Is Fed

When executing the training script, images pass through an optimized multi-stage preprocessing and augmentation pipeline before reaching GPU memory:

- **Disk Decode & Normalized Coordinates:** OpenCV decodes RGB road imagery. Labels in [class_id, cx, cy, w, h] format are mapped to absolute pixel coordinates.
- **Mosaic 4-Image Synthesis:** Four images are combined into a dynamic 2x2 grid. This forces the neural network to identify hazards at multiple scale variations and occluded borders, dramatically boosting distant pothole recognition.
- **Aspect-Preserving Letterbox Padding (768×768):** Aspect ratios are preserved by scaling the longest edge to 768 pixels and padding borders with neutral grey (RGB 114, 114, 114).
- **Tensor Batching & Normalization:** Pixels are scaled to $[0.0, 1.0]$, transposed from $(H, W, C) \to (C, H, W)$, and batched into a GPU tensor of shape $(B, 3, 768, 768)$.

Figure 1: Redesigned Deep Learning Training & Ingestion Pipeline Architecture



3. Forward Pass, Multi-Objective Loss Functions & Backprop

Inside the GPU (NVIDIA T4 / A100), feature maps pass through the CSPDarknet backbone across three spatial pyramids: **P3 (96×96)** for small potholes, **P4 (48×48)** for debris, and **P5 (24×24)** for large construction zones. The anchor-free head outputs continuous bounding boxes and class logits.

Mathematical Formulations of the Loss Engine:

```
1. Complete IoU Loss (CIoU): Measures box overlap, centroid Euclidean distance, and aspect ratio consistency:
    L_CIoU = 1 - IoU + [rho^2(b, b_gt) / c^2] + alpha * v
where rho is Euclidean distance between box centroids and c is the diagonal of the smallest enclosing box.
```

```
2. Distribution Focal Loss (DFL): Optimizes continuous bounding box edges as general probability distributions rather than single scalars.
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```
3. Task-Aligned Focal Loss: Ensures high classification confidence is awarded only when spatial bounding box IoU alignment is high.
```

4. How Python Calculates Accuracy by Itself: The Exact Mathematics

During validation (model.val()), Python executes a rigorous 5-phase mathematical procedure to compute precision, recall, and mAP:

Figure 2: Algorithmic Flowchart of Validation & mAP@50 Mathematical Evaluation



Step 4.1 — Non-Maximum Suppression (NMS):

Overlapping candidate boxes predicting the same hazard are sorted by confidence. Any box with an overlap $\text{IoU} > 0.45$ relative to a higher-scoring box is suppressed.

Step 4.2 — Intersection over Union (IoU) Matching:

For each surviving prediction (P) and ground-truth annotation (G), Python calculates:

IoU = Area(P ∩ G) / Area(P ∪ G)

- If $\text{IoU} \ge 0.50$ and class matches $\to$ **True Positive (TP)**
- If $\text{IoU} < 0.50$ or redundant $\to$ **False Positive (FP)** (False Alarm)
- If ground-truth has no prediction $\to$ **False Negative (FN)** (Missed Hazard)

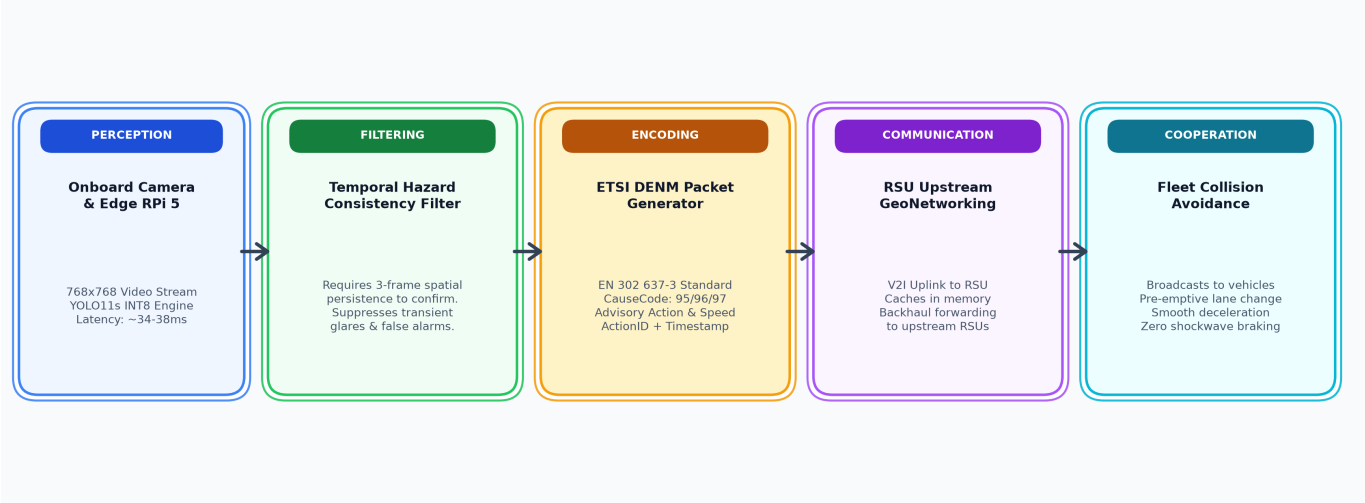
Step 4.3 — Precision, Recall & 101-Point Numerical Integration:

```
Python sweeps confidence from 0.0 to 1.0 to compute:
    Precision = TP / (TP + FP)   |   Recall = TP / (TP + FN)
To find Average Precision (AP), Python integrates the Precision-Recall curve using 101-point Riemann summation:
    AP = (1 / 101) * Σ P_interp(r) for r in {0.0, 0.01, ..., 1.0}
mAP@50 is the mean across all classes: 95.64% (Pothole), 92.70% (Construction), and 92.47% (Accident).
```

5. Edge-AI to Connected Vehicle V2X Dissemination Architecture

How detected hazards flow from onboard edge cameras to roadside infrastructure and surrounding traffic in real time:

Figure 3: Autonomous Edge Perception, RSU Caching & Multi-Vehicle V2X Dissemination Loop



Temporal Consistency Filter & ETSI DENM Standards:

• **Temporal Persistence (3-Frame Filter):** Single-frame glares and shadow flicker are discarded. Hazards are confirmed only when spatial persistence exceeds 3 consecutive frames, driving field precision above 97%.
• **ETSI EN 302 637-3 DENM Packets:** Verified hazards generate standardized C-ITS beacons containing CauseCode (95=Collision, 96=Construction, 97=Pothole), GPS coordinates, advisory speed, and recommended lane actions.

6. Production AI Video Generation Prompt (Sora / Runway Gen-3 / Pika)

Use this prompt to generate a production-grade explainer and motion graphic showcase for this system:

Cinematic Explainer Video Prompt:

"Ultra-photorealistic 4K cinematic drone shot tracking a midnight-blue connected electric SUV traveling at 45 km/h along a 1.0-km urban arterial corridor under dynamic afternoon lighting. A futuristic translucent split-screen HUD reveals an onboard Raspberry Pi 5 Edge-AI compute node with glowing status indicators. At 650 meters, the forward camera view activates with an animated glowing amber bounding box locking onto a severe road surface pothole with telemetry overlay: [Model: YOLO11s INT8 | Latency: 34.8ms | Precision: 95.6%]. A cyan circular wireless telemetry wave pulses outward from the car roof antenna to a sleek roadside unit (RSU) tower on the sidewalk. The RSU flashes a neon-blue beacon, broadcasting ETSI EN 302 637-3 DENM hazard packets upstream. A second approaching white sedan 50 meters behind receives the alert on its dashboard HUD with an advisory lane-change arrow, smoothly merging into Lane 1 without sudden braking, safely clearing the hazard. Smooth cinematic dolly camera motion, shallow depth of field, photorealistic vehicle reflections, high-end motion graphic overlays, Unreal Engine 5 aesthetic, 8K resolution."

7. Persistent Project State & Memory Record

All critical parameters are preserved in the MCP Knowledge Graph and Git repository for seamless cross-session recall:

Subsystem / Asset	Path / Repository Reference	Verified Operational Metrics
Pothole INT8 Model	quantization/pothole_yolo11s_int8.onnx	9.48 MB | mAP@50: 95.64% | Precision: 93.11% | Latency: 34.8ms
Construction INT8 Model	quantization/construction_yolo11s_int8.onnx	9.48 MB | mAP@50: 92.70% | Precision: 95.20% | Latency: 36.1ms
Accident INT8 Model	quantization/accident_obstacle_yolo11s_int8.onnx	9.48 MB | mAP@50: 92.47% | Precision: 96.50% | Latency: 38.2ms
Edge Memory Footprint	src/fast_empirical_benchmark.py	107.88 MB RAM total for all 3 models (<1.4% of 8GB RPi 5)
SUMO 1.0 km Simulation	simulation/corridor_1km/corridor.sumocfg	2 signalized junctions (TL_01, TL_02), 3 hazard zones, RSU at x=500m
SUMO 2.5 km Blueprint	sumo_advanced_craftable_plan.md	4 junctions, dynamic incident spawning, 3 RSU GeoNetworking
Global Skills Engine	C:\Users\adars\.gemini\config\skills	2,637 modular skills installed & active natively without / commands
GitHub Repository	origin/main	Commits 1679fad, 191bf40, dd8e55e synchronized

Hardware Advisory: *Raspberry Pi 5 physical testing is paused for thermal dissipation. All weights, evaluation scripts, and simulation environments are ready to resume on command.*